I. Overview

1. The Coupled MSD System Model and Dynamics

The plant under control is a Coupled Mass-Spring-Damper (CMSD) system, a 4-state mechatronic architecture representative of active suspension systems or dual-mass industrial positioning stages. The system consists of two masses (m_1 and m_2), connected to each other and to a fixed reference frame by springs (k_1, k_2) and dampers (b_1, b_2) The control effort $u(t)$ is applied directly to m_1, making this a non-collocated control problem where the primary variable of interest (the position of m_2) is physically separated from the actuator.

The Transfer Function:

To analyze the system in the frequency domain, we derive the transfer function $G(s) = \frac{X_2(s)}{U(s)}$

Applying Laplace transforms to the differential equations of motion yields:

$$m_1 s^2 X_1(s) + (b_1 + b_2)s\, X_1(s) + (k_1 + k_2)X_1(s) - b_2 s\, X_2(s) - k_2 X_2(s) = U(s)$$

$$m_2 s^2 X_2(s) + b_2 s\, X_2(s) + k_2 X_2(s) - b_2 s\, X_1(s) - k_2 X_1(s) = 0$$

Solving for $X_2(s)$relative to $U(s)$ gives the general transfer function:

$$G(s) = \frac{(b_2 s + k_2)}{[(m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2)][m_2 s^2 + b_2 s + k_2) - (b_2 s + k_2)^2]}$$

The 4th-order denominator is the complex, resonant internal dynamics that the controller must manage to achieve stable tracking.

2. Control Methodologies

To evaluate optimal performance, six distinct control architectures were implemented and tested:

Proportional-Integral-Derivative (PID) with Feedforward:

- *Definition:* A model-free controller that calculates effort based on the present (P), past (I), and predicted future (D) errors.
- *Pros:* Simple to tune, universally understood, highly precise for static loads.
- *Cons:* Reactive rather than predictive; entirely blind to the internal physics (like the spring resonances) of the CMSD system.
- *Note: Feedforward Mechanism:* A feedforward term calculates and injects the exact baseline force required to hold a position (e.g., fighting gravity or steady-state spring tension), leaving the PID feedback loop responsible only for correcting transient errors.

Adaptive PID:

- *Definition:* A PID controller where the , and gains dynamically shift based on operating conditions or error thresholds.
- *Pros:* Can adjust to varying payloads (e.g., changes to).
- *Cons:* High computational overhead; can become unstable if the adaptation mechanism overreacts.

Linear Quadratic Regulator (LQR):

- *Definition:* An optimal, model-based state-feedback controller. It uses a cost function () to find the perfect mathematical balance between state errors (Q matrix) and energy usage (R matrix).
- *Pros:* Highly efficient; mathematically guarantees stability by observing all 4 states simultaneously.
- *Cons:* Without an integral term, it acts purely as a regulator (driving to zero) and struggles with non-zero tracking.

PID + LQR Hybrid:

- *Definition:* A cascaded architecture where LQR handles the internal stabilization (inner loop) and PID forces the setpoint tracking (outer loop).
- *Pros:* Best of both worlds conceptually.
- *Cons:* Complex to tune; the two loops can "fight" each other, leading to high energy spikes.

Sliding Mode Control (SMC):

- *Definition:* A robust, non-linear controller that defines a "sliding surface" in the state space and uses high-speed switching to force the system onto that surface, rendering it immune to matched disturbances.
- *Pros:* Extreme robustness; theoretically perfect tracking even with unknown payloads or heavy disturbances.
- *Cons:* "Chattering" (high-frequency vibrations) which can damage mechanical actuators if not smoothed by a boundary layer.

Adaptive SMC:

- *Definition:* An SMC where the switching gain (η) or the boundary layer (ϕ) adapts in real-time. In this case, our algorithm is based on the Culture-Bees Algorithm, where α, λ, μ (corresponding to η, C, ϕ) are re-optimized at every 0.01s sample by running the full Culture-Bees search.
- *Pros:* Maximizes precision when far from the target while aggressively smoothing the signal near the target to kill chattering.

Model Predictive Control (MPC):

- *Definition:* A control strategy that uses a mathematical model of the system to predict future behavior over a specific time horizon. At each time step, it solves an online optimization problem to find the best sequence of control inputs that minimizes a cost function while strictly respecting system constraints (e.g., motor torque limits or maximum velocity).
- *Pros:* Inherently handles multi-variable systems and complex constraints; provides "look-ahead" capability to preemptively adjust for upcoming setpoint changes or known disturbances, resulting in highly efficient and smooth trajectories.
- *Cons:* Computationally expensive, as it requires solving an optimization algorithm in real-time at every sampling interval. Furthermore, performance is heavily dependent on the accuracy of the internal system model.

3. Evaluation Model

Performance is judged on two primary vectors: Tracking Precision (x_2) and Actuator Wear/Energy (u). The main metrics used are:

Mean Absolute Error (MAE): The average absolute deviation from the target. Used to define the overall accuracy of x_2 and the average continuous power draw of u.

Formula:

$$MAE = \frac{1}{n}\sum |Actual - Target|$$

Root Mean Squared Error (RMSE): Squares the error before averaging, heavily penalizing large deviations. Applied to x_2, it highlights unwanted oscillations/bouncing. Applied to u, it highlights violent, high-frequency motor spikes that cause mechanical fatigue.

Formula:

$$MSE = \frac{1}{n}\sum (Actual - Target)^2$$

$$RMSE = \sqrt{MSE}$$

II. Simulink models

The Coupled MSD System Model

The CMSD subsystem was constructed in Simulink to represent the active suspension of a heavy vehicle.

Unsprung Mass (Wheel/Axle): $m_1 = 250\ kg$, $k_1 = 750\ N/m$, $b_1 = 50\ Ns/m$

Sprung Mass (Chassis/Load): $m_2 = 300\ kg$, $k_2 = 250\ N/m$, $b_2 = 100\ Ns/m$

The control input (u) acts directly on m_1, simulating an actuator placed between the ground and the chassis. This creates a "non-collocated control" problem, as the primary variable of interest (x_2) is separated from the actuator by a flexible spring (k_2).

All systems are fitted with a "Band-limited White noise" block with noise power 200 (N), 0.01s sampling time and seed: 23341. And the Control input power is capped at [-3500; 3500] (N).

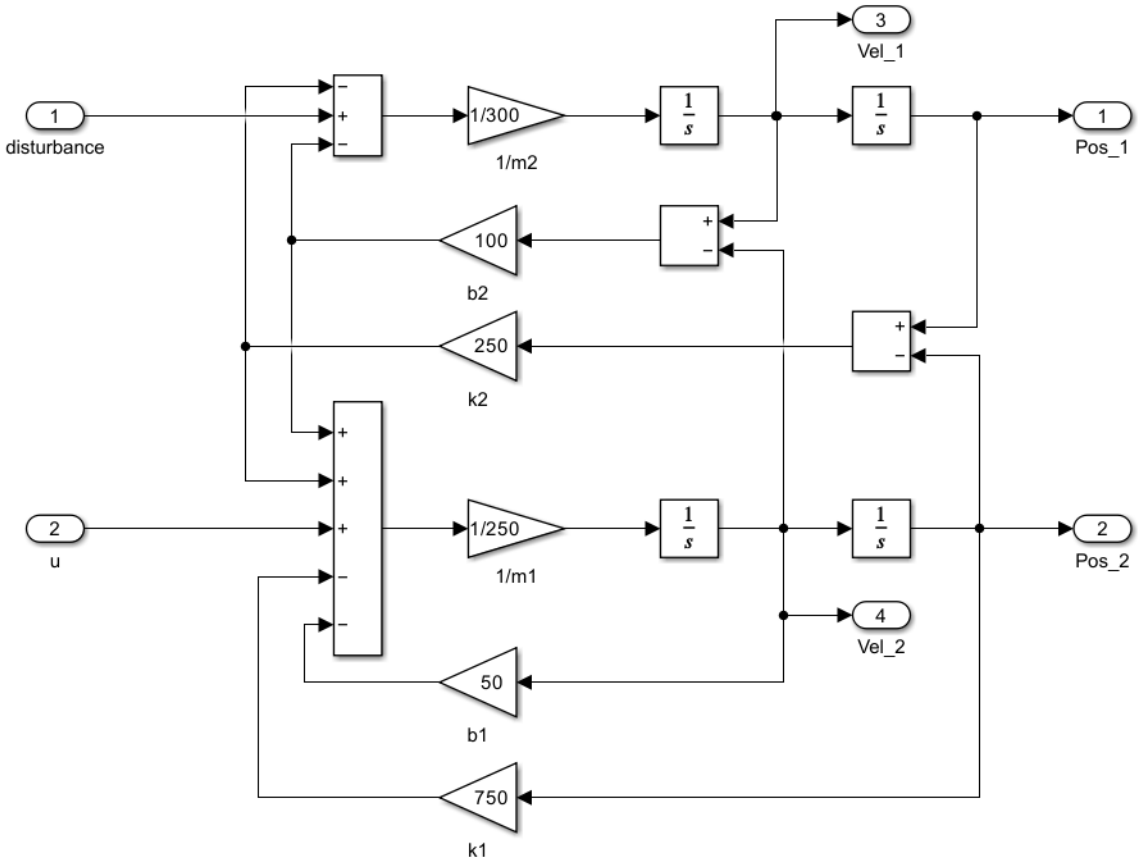


Figure 2.1: Simulink model of the Coupled Mass Spring Damper system

III. Simulations

1. PID with Feedforward

a. Model

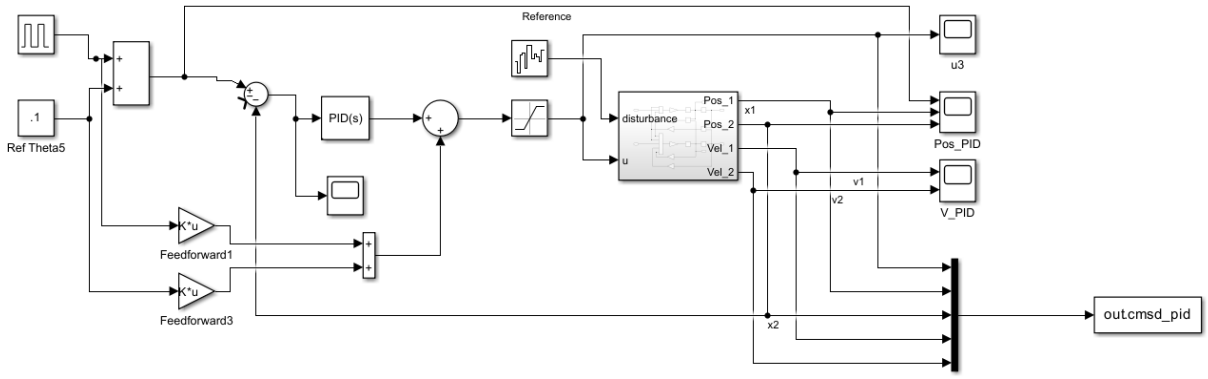


Figure 3.1: PID Control model for the CMSD system

The Controller's parameters are as follow: $K_P = 12000; K_I = 1100; K_D = 5000$. With Feedforward Gains for the step command and the original 0.1 reference points being 200 and 300 respectively.

b. Result Diagrams

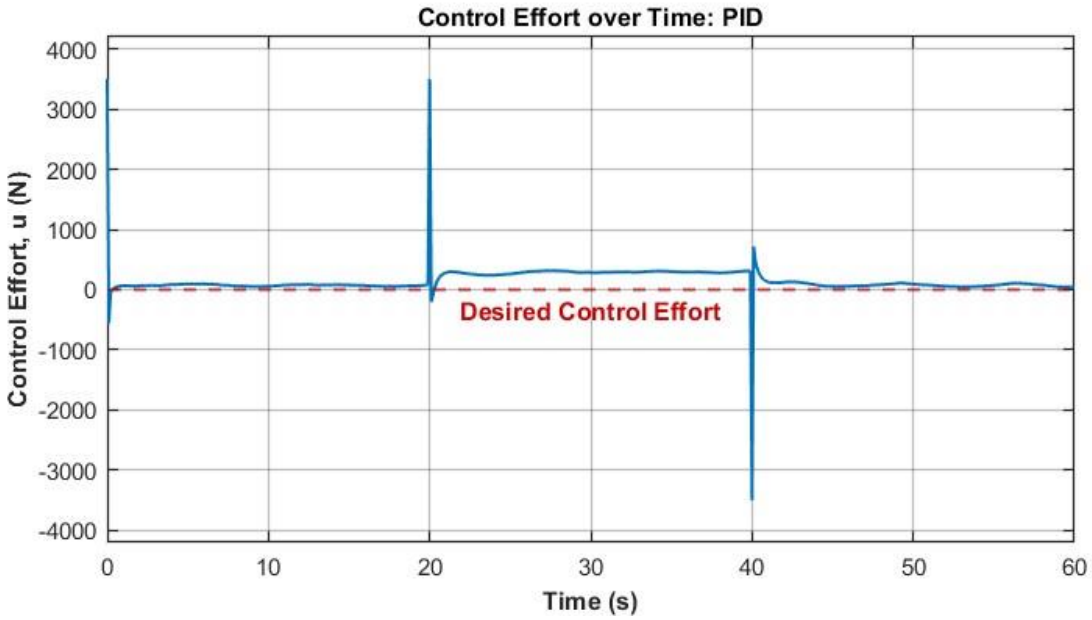


Figure 3.2: PID Control's Effort over time

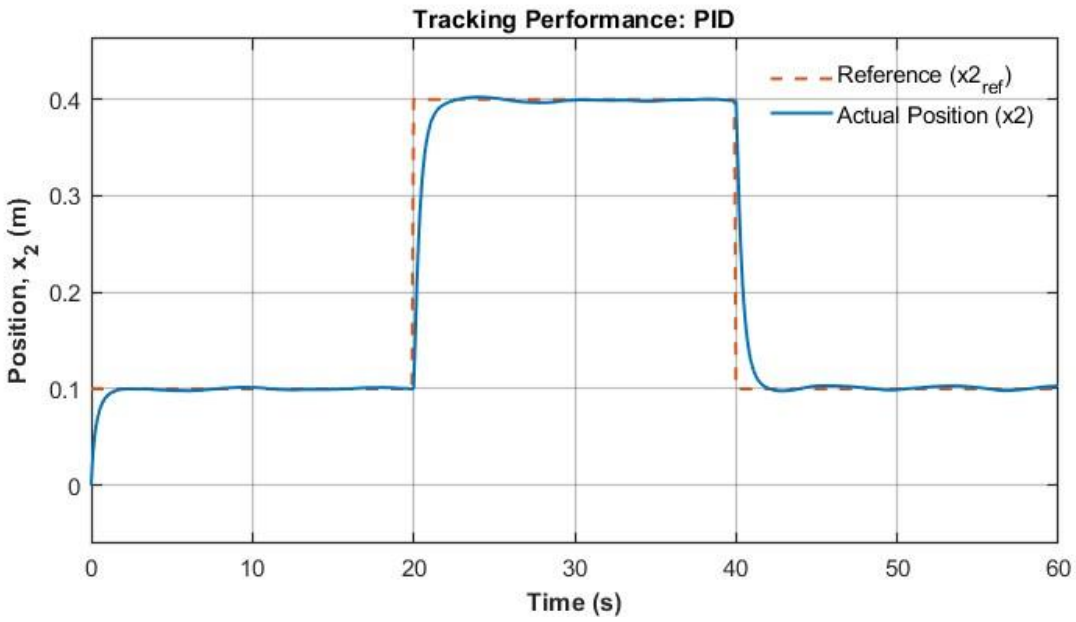


Figure 3.3: PID Tracking performance over time

c. Numerical Results & Verdict

Tracking (x_2): MAE = 0.0064 m | RMSE = 0.0288 m
Effort (u): MAE = 165.04 N | RMSE = 307.16 N

Verdict: Standard PID achieves near-perfect tracking precision and rise time but is prone to disturbance as seen in the graph. However, due to it lacking knowledge of the system's physics, it forces the error to zero by violently acting on the actuator, resulting in the highest mechanical stress of all tested controllers.

2. SMC

a. Model

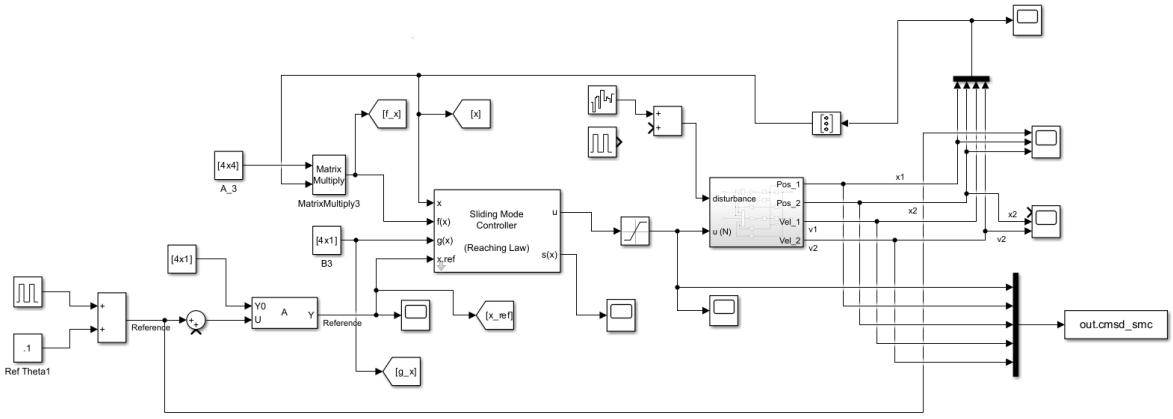


Figure 3.4: SMC model for the CMSD system

The Controller's parameters are as follows:

Sliding Coefficient Matrix: [0; 8; 0; 4]

Exponential Reaching Law: $\eta = 20; K = 10$

Hyperbolic Tangent boundary layer: $\phi = 0.0274$

b. Result Diagrams

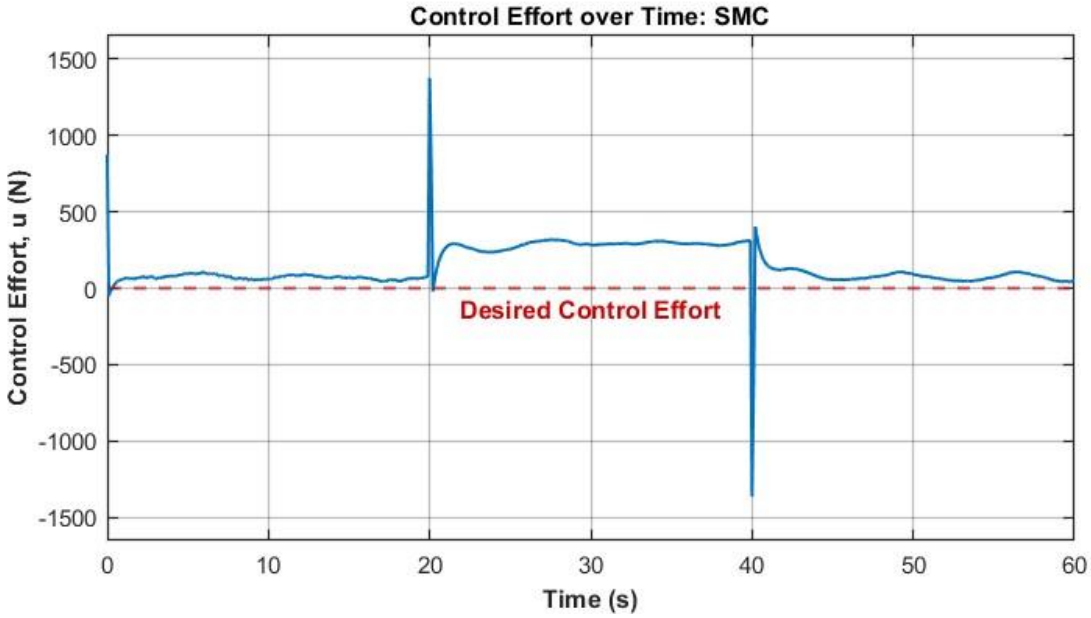


Figure 3.5: SMC's Effort over time

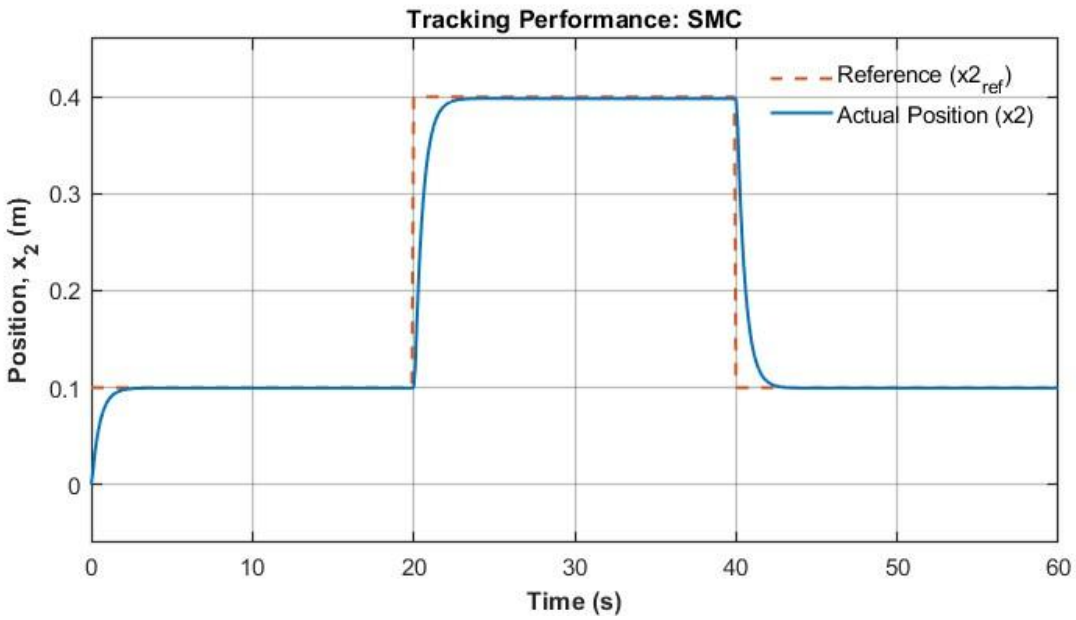


Figure 3.6: SMC Tracking performance over time

c. Numerical Results & Verdict

Tracking (x_2): MAE = 0.0077 m | RMSE = 0.0336 m
Effort (u): MAE = 153.00 N | RMSE = 201.33 N

Verdict: SMC proves to be highly resilient to disturbances as theoretical models suggest. By utilizing the A and B state-space matrices, it provides tracking precision that is 95% as accurate as pure PID, but requires roughly 34% less peak mechanical force (RMSE 201 vs 307) to achieve it.

3. Adaptive SMC

a. Model

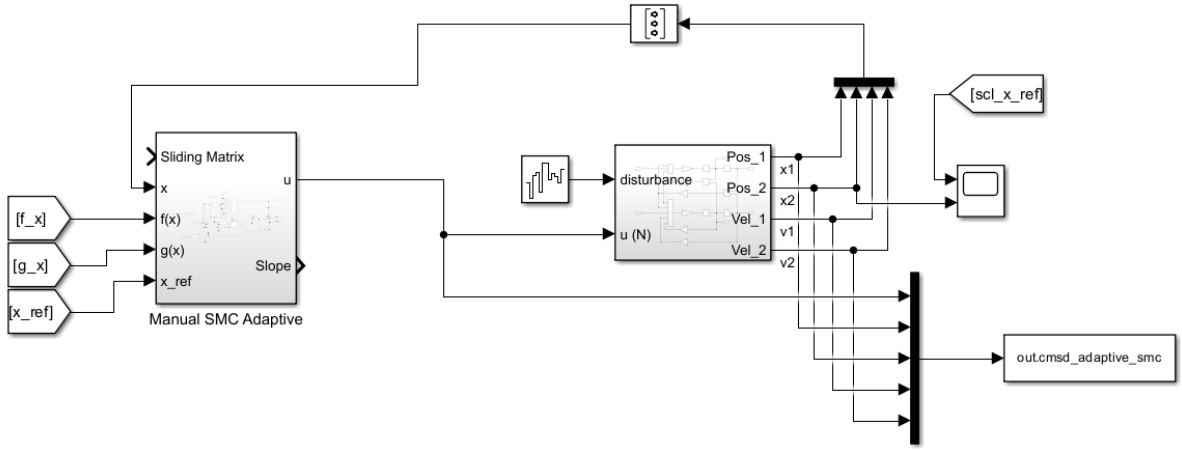


Figure 3.7: Adaptive SMC model for the CMSD system

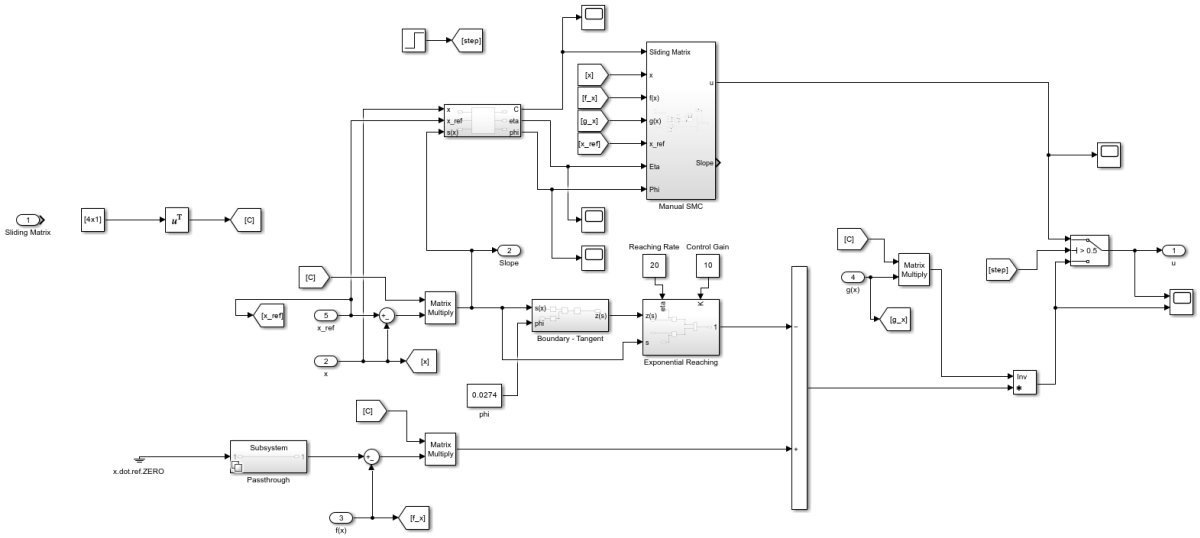


Figure 3.8: Adaptive SMC internal view

The Controller's parameters are as follows:

Sliding Coefficient Matrix: [0; 3÷15; 0; 3÷15]

Exponential Reaching Law: $\eta = 8 \div 25; K = 10$

Hyperbolic Tangent boundary layer: $\phi = 0.02 \div 0.1$

b. Result Diagrams

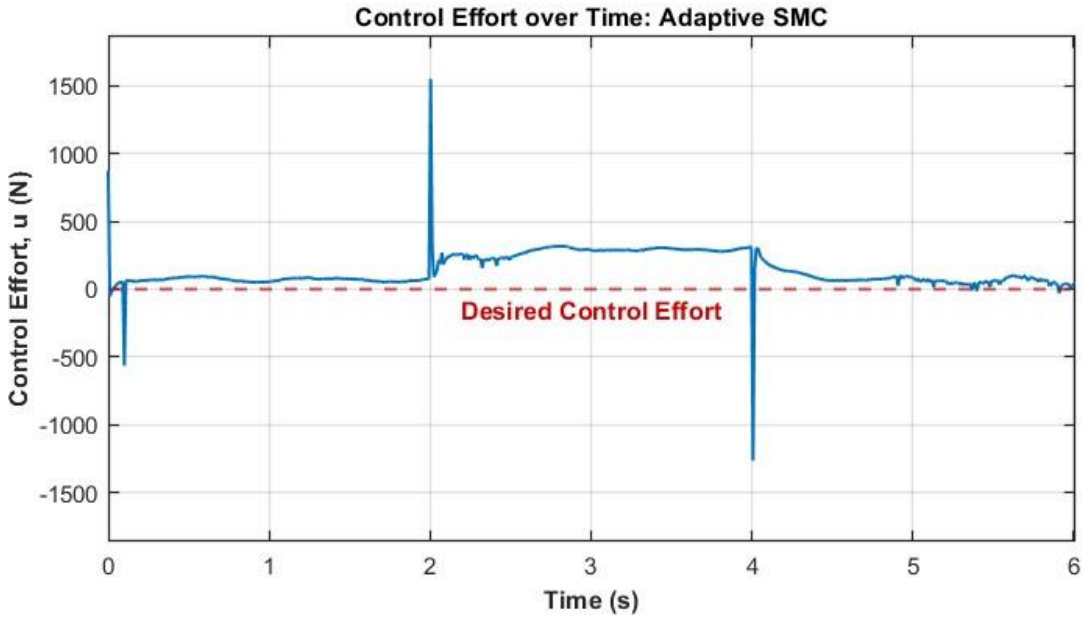


Figure 3.9: Adaptive SMC Control's Effort over time

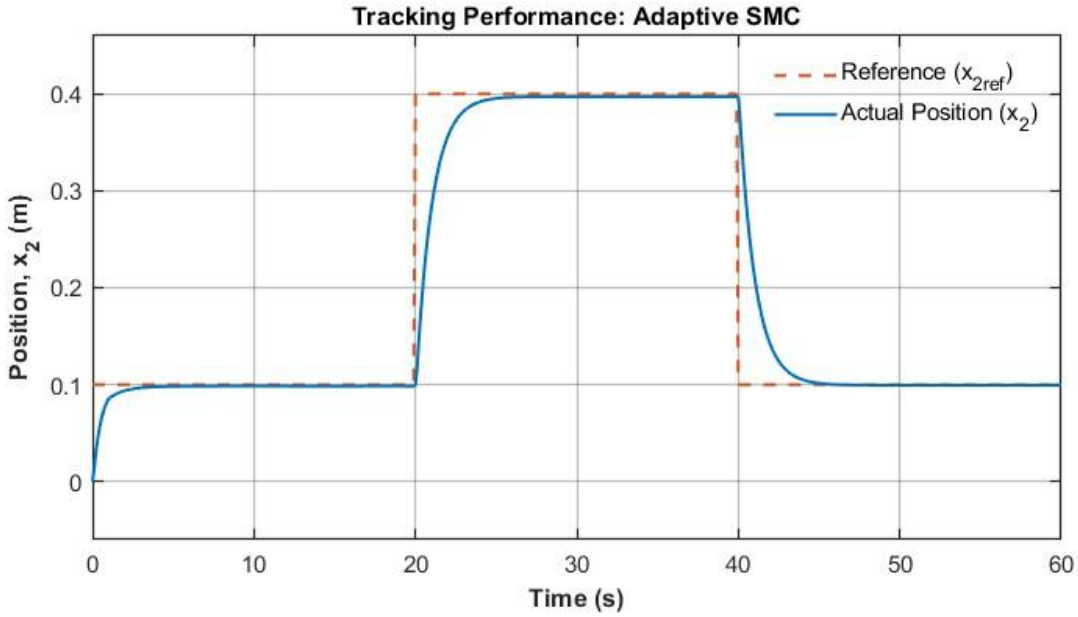


Figure 3.10: Adaptive SMC Tracking performance over time

c. Numerical Results & Verdict

Tracking (x_2): MAE = 0.0133 m | RMSE = 0.0425 m
Effort (u): MAE = 147.89 N | RMSE = 196.48 N

Verdict: Adaptive SMC achieves the second-lowest control effort (RMSE 196.48 N), confirming that the adaptive gain law successfully reduces unnecessary switching near the sliding surface. However, this gain reduction comes at a tracking cost: positional error is 26% higher than plain SMC (RMSE 0.043 vs 0.034 m). The adaptation parameters require further tuning to recover tracking precision without sacrificing the effort savings.

4. Model Predictive Control

a. Model

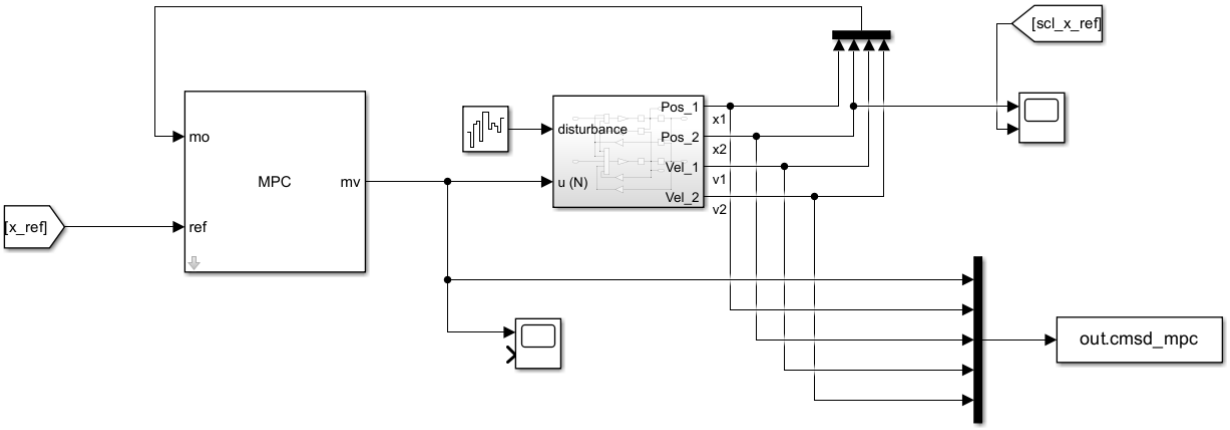


Figure 3.11: MPC model for the CMSD system

The Controller's parameters are as follows:

Sample time: 0.01; Prediction Horizon = 50; Control Horizon = 5

Constraints (for u): $u \in [-3500; 3500]; rate\ change\ \leq 500N/step$

Weight: $[x_1; x_2; v_1; v_2] = [1; 1e5; 5; 10]$

b. Result Diagrams

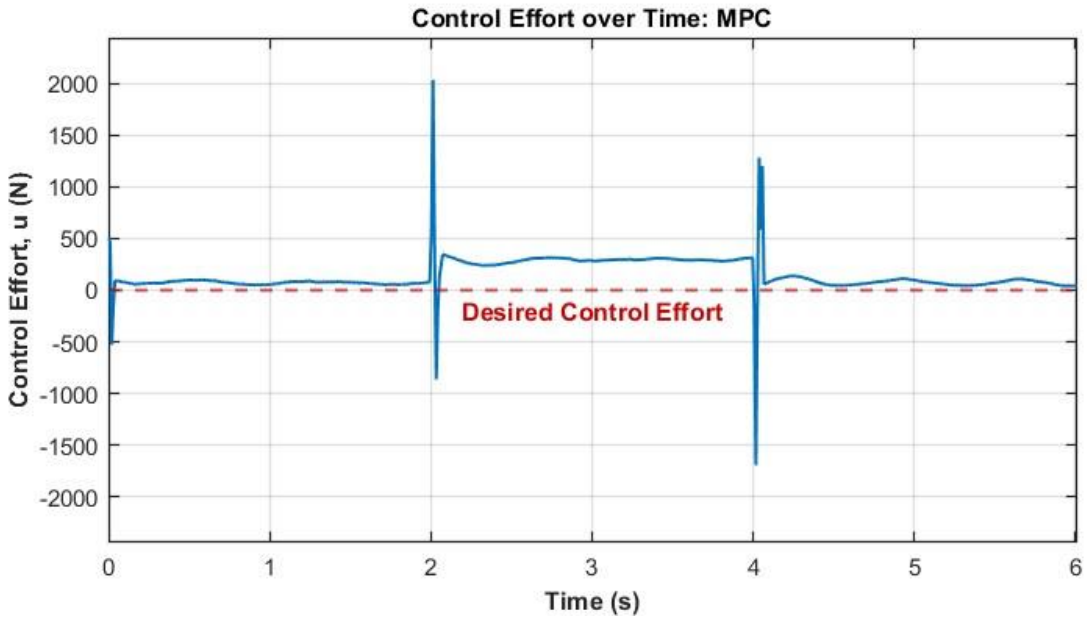


Figure 3.12: MPC's Effort over time

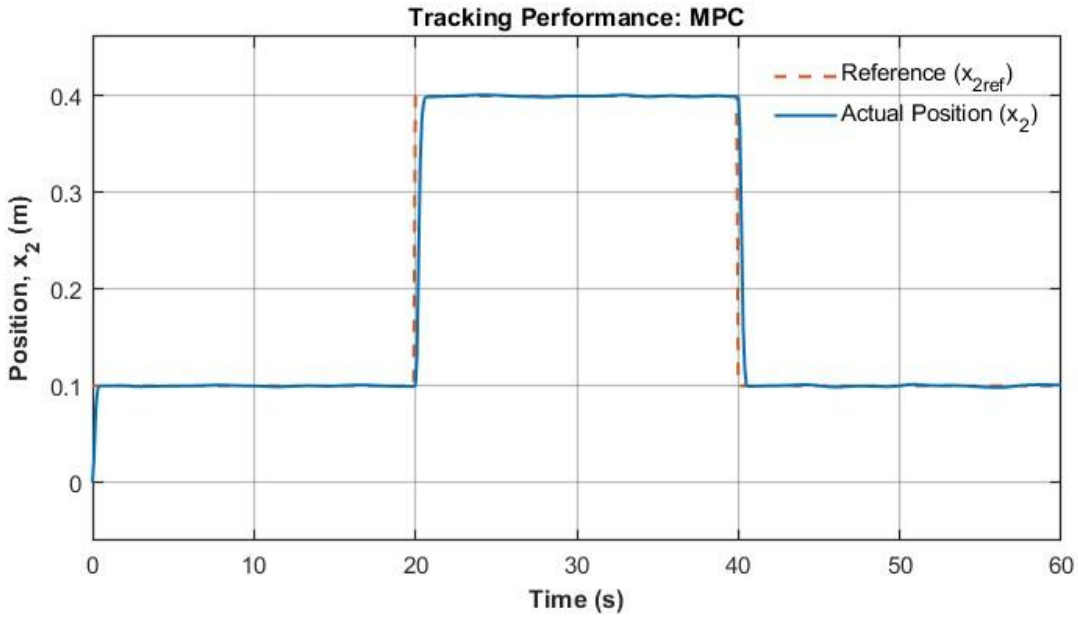


Figure 3.10: MPC Tracking performance over time

Tracking (x_2): MAE = 0.0036 m | RMSE = 0.0259 m
Effort (u): MAE = 160.23 N | RMSE = 228.01 N

Verdict: MPC delivers the best tracking performance of all controllers tested, with the lowest MAE (0.0036 m) and RMSE (0.0259 m) for position x_2. Its predictive horizon allows it to anticipate reference changes rather than react to them. This precision comes at a moderate effort cost (RMSE 228 N), 16%

more than SMC but with 23% better tracking accuracy, a favourable trade-off for applications where positional precision is the primary objective.